

IoT-based Smart Water Quality Monitoring System for Malaysia

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Abstract – Rapid development has also negatively impacted the sources that provide raw water for agricultural, domestic, and industrial needs. All these sources have been polluted due to man's pure pursuit of economic greed, which has caused illegal waste dumping and high amounts of industrial waste above government regulations, surpassing the ecosystem's self-purification capabilities. In addition, these water systems lead to treatment plants across the country to be treated before daily consumption for domestic use. There have been cases whereby even the treatment plants had to be shut down, which caused water disruptions, namely in Selangor, Kuala Lumpur, and their neighbouring regions. Water quality today should be at its best; however, there have been many occurrences when it has been proven otherwise, both in Malaysia and internationally. In this paper, the authors have proposed an IoT based solution that helps people easily monitor the water quality supplied to their homes.

Keywords – Water Quality, Water Purification, Water Treatment Plant, IoT-based Water Quality Monitoring

I. INTRODUCTION

In recent years, the quality of water that we are using to shower, cook food and almost everything in between. Even though it is vital, people seem to disregard the importance of it, which is naive. In 2020, Malaysians saw numerous water cuts in the capital, Kuala Lumpur and her neighbouring cities, mainly in July and December, caused by human dumping rivers leading to the water treatment Centre [1]. The utility company was able to handle it in a very professional manner. Still, many people complained that they had been tasting and smelling something in the water for at least two weeks before the utility company took any action [2].

That was not an isolated event, as we saw the same thing happen this year almost every month. We only managed to catch one factory taken to court by the state and shut down even though the utility company and authorities acted quickly. What about the many other factories that have this unethical practice [3]. This shows laxity on the authority's part, and the utility company for not protecting it is a water source [4].

Other than that, there have been many cases where the water was polluted after it had gone through the utility companies treatment centre and was in good quality but was in turn contaminated on the way to its users. The most prominent example of this was when it happened in Flint, Michigan, in the United States of America, where a lead from the pipes leaked into the water supply and caused many people to get sick, namely children. The affected of lead poisoning was sadly only confirmed at a late stage and had already affected many people. The cause was the utility company purposefully putting in more corrosive chemicals to cut corners and save cost in cleaning the water for human

consumption. This has affected people in Flint up to today, which marks six years of not having a safe water source [5].

To analyse, implement, and design a system that connects to an android application using IoT to monitor the quality of water supplied to homes across Malaysia. Based on the problems stated above, a system with IoT capabilities that connects to a mobile application should be implemented to monitor water quality entering homes. Implementing this system will allow residents to check and balance with utility companies and help them sleep better at night. It knows that they can monitor the water quality being used, especially if there are children in their household, as they are more susceptible to contaminants like lead that can be detrimental to their wellbeing. Besides, this system will also enable the utility company to gather data if the user is willing to share it and maintain their water supply systems as there could be defects in the pipes causing contaminants to seep into the water supply of a particular area. This would help them roughly locate and identify which site to focus on that may need upgrading. Other than that, this will act as proof that something may be contaminating the water supply. When residents complain to utility companies, they will be able to act faster in the case of contamination and be notified immediately of the proof of contaminants in the water.

II. PROBLEM STATEMENT

The problem mainly starts when we depend on utility companies to provide a safe water supply that may only be tested before leaving the water treatment facility. As a result, pollutants could affect the water quality after it leaves the treatment facility. An example of this occurring was in Flint, Michigan, the water leaving the treatment facility had an excess of chemicals. In addition, the city pipes were lead pipes, which caused the water to erode the pipeline, causing lead to dissolve into the water. These caused many people to get sick with lead poisoning, which significantly affected kids[6].

Another example is when Selangor and its neighbouring regions frequently face water cuts due to traces of pollutants in the water. They were identified quickly at the water treatment facility, which is not an issue. When the contaminants were detected late, and residents used the polluted water for their daily intake. This happened in 2017 when the water supply was disrupted for almost a week in some parts, and it is made extremely difficult for people to just depend on water quality monitoring to be done by utility companies in their treatment facility, which cannot guarantee the same quality of water, leaving the facility is entering the water supply system [7].

Implement a water monitoring system into all homes so that every resident knows the water quality that they are supplied is in good order. Furthermore, water utility companies may benefit from the data gathered and improve their services with the data mainly based on the areas that may have pollutants seeping into the water whilst going to the user. Therefore, an appropriate system should be developed and implemented to help residents easily monitor water quality and collect data from utility companies.

The proposed system can provide the water quality readings in real-time when the user opens the application on an android device and will be user-friendly as possible. The proposed system will also be able to send an email with the history log of sensor readings to users email when requested and send the user an email with a template and history of sensor readings to be sent by the user to the utility company. The proposed system will provide the following functionalities to the user:

- Hardware can communicate with the server with WiFi.
- Login to both applications that will be connected to a database.
- Display a dashboard with real-time water quality readings.
- Show the option to see the history of water readings.
- Send the historical reading to the application.
- Show utility company's contact within the application.
- Show information regarding safe water quality and the meaning of sensor readings.
- Contact admin for support.

To implement such an elaborate system for public use, the research itself is a challenge as this project requires in-depth knowledge of the subject matter. With that, one of the first challenges would be researching what hardware is needed, and this would need an understanding of components such as sensors and other such hardware required to make the components work and protect them. Extensive research would need to be conducted on the system software, communication application Components to a centralised server.

Another area that will need extensive research would be the ideal connectivity method. Other than that, challengea would be faced ensuring this system can connect to the server in any environment as it will be used in many different locations and types of houses. For this, the researchers should implement 'Extended Coverage GSM IoT' or widely known as EC- GSM- IoT, basically Low Power Wide Area (LPWA) technology is based on GPRS and designed as a high capacity, long-range, low energy, and low complexity cellular system for IoT communications. However, the researchers will also need to consider the costs [6]. Another challenge would be to protect the hardware of the system as it will be placed near the water meter that may be placed in many different environments based on the type of home the user is living in, the researchers would have to take into account that some of the hardware cannot come into contact with water. In contrast, some components, such as the sensors, would need to contact the water. The researchers will also need to take into consideration the immense pressure that the encapsulation may be under from the weather as well as the water, as the service water pipe has a standard pressure for residential properties ranging between 30 pounds per square inch (psi) to 45 psi [7].

Another challenge that the developers may face is ensuring the system's security as a whole; this would include physical attacks, network attacks, software attacks and encryption attacks that could jeopardise the system or the privacy of a user. Thus, the developers will need to have done good research and understand how these attacks occur to prevent them from happening [10].

III. THEORETICAL FOUNDATIONS OF THE STUDY

The main problem is the water quality that is being supplied to homes in Malaysia. With the frequent water cuts due to the pollution of water sources, residents should act as a check and balance to the utility companies water quality tests and prevent late reactions from utility companies when there is sub-par water quality. This study will investigate the primary factors and existing solutions, comparing them and gaining insight as to how to improve the proposed system.

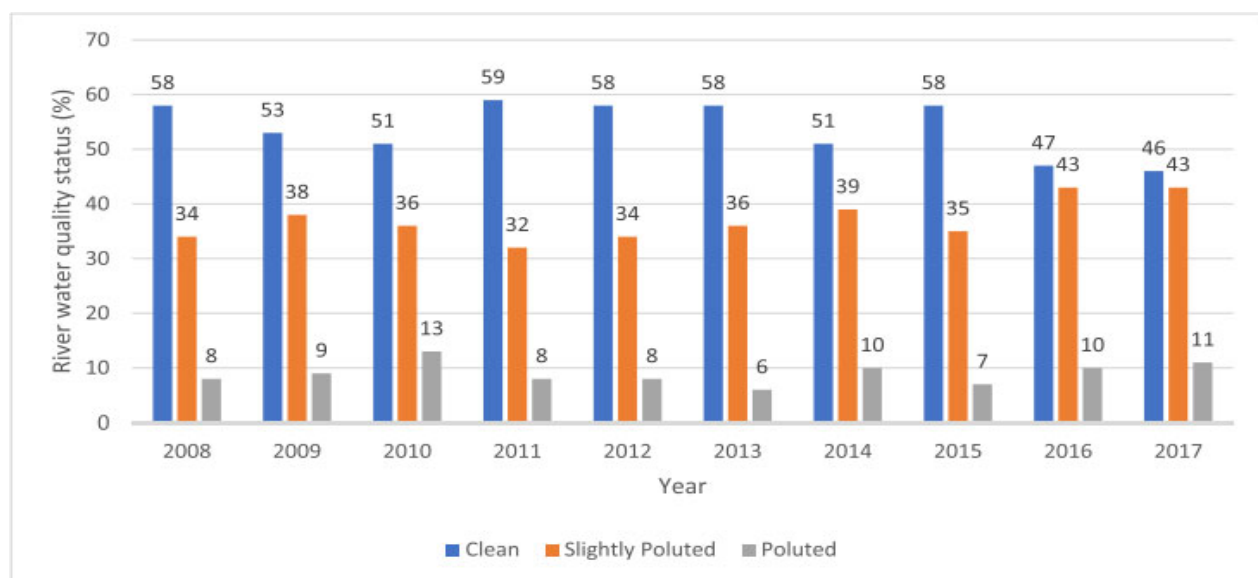


Fig. 1. Quality of river water 2008 – 2017 [10]

A. Issues Affecting Water Quality

Malaysia is a country with an abundance of water; 98% of the total water consumed originates from these rivers. To be exact, this includes water for agriculture, industrial use, and domestic use. Therefore, as a developing country seeing rapid growth of population and rapid urbanisation, this has led to a higher demand for water [11]. That said, rivers in Malaysia directly impact the water quality being supplied to homes across Malaysia. In Fig. 1, we see the trend of water quality of rivers in Malaysia from 2008 to 2017, which sees a significant drop in clean water [12].

This may not be all that safe caused by the rapid developments as the river quality in Malaysia is deteriorating; this is important because almost. With that, an easy conclusion can be drawn that these rivers directly impact the quality of water that is being supplied to homes [12]. Fig. 1 obtained from the Department of Environment shows the river quality trend from 2008 to 2017. It lessens the amount of clean water left in the rivers from 2008 to 2017; this has also caused the number of waterways to decrease from 579 rivers in 2008 to 477 in 2017 [18].

As shown in Fig. 1, water pollution in recent years have become worse, and the conventional treatment systems which are widely used still are not able to treat such complex water characteristics with the presence of new contaminants and heavy metals; the traditional model of treatment is fundamental it includes screening, coagulation and flocculation, sand filtration, chemical disinfection which mainly is chlorine, and fluoridation. This process is illustrated in Fig. 2 [12].



Fig. 2. Conventional water treatment model [10]

Although conventional water treatment systems are not well suited in Malaysia due to heavy pollutants in the water, they are still used in most treatment plants. Whereas the more advanced non-conventional method such as Activflo Clarification System, Dissolved Air Flotation (DAF) and Ozone, Ultra Membrane Filtration is only used in a small number of plants, despite knowing the usage of such technology can improve the water [19]. This means there is still the presence of heavy metals in the water that is supplied to consumers across Malaysia. According to World Health Organisation (WHO), these are the heavy metals that can leach into the water, chromium (Cr), cadmium (Cd), zinc (Zn), copper (Cu), nickel (Ni), lead (Pb), aluminium (Al), iron (Fe), arsenic (As). However, the Malaysian Ministry of Health (MOH) has listed heavy metal parameters with National Drinking Water Quality Standard that is to be strictly adhered to by treatment facilities. These metals can still be leached into the water by corroded pipes [19].

B. Factors Affecting Tap Water Quality

The heavy metals above can get into the water by either a corroded pipeline or from the water source due to pollution, natural occurrences, and anthropogenic activities. Although most pipes that are known to cause these leaches have been upgraded to piping that is less likely to cause the leeches, there are still pipes that have not been changed. Fe, As, and Al metals naturally occur in soils that have been known to infiltrate into the water body, but these metals are easily

removed through treatment methods widely used in Malaysia. The anthropogenic activities that cause heavy metals to get into water bodies are mining, industrial, and industrial activities [19].

Agricultural activities contribute to pollution by inadequate or improper waste management and runoff from fertiliser. Still, we recently saw treatments centres across Selangor shut down due to a duck farm clean-up that had waste enter the Sungai Semenyih and Sungai Langat; these rivers are the primary source of water for a vast section of Selangor. In some cases, we also see pollution due to the illegal dumping of chemicals, which saw water disruptions across Kuala Lumpur in 2019 and 2020 [19].

These cases are not isolated as they have been happening for years and are occurring more frequently in the past few years, causing water disruptions every few months. In some cases, the utility companies had acted late, causing consumers to lodge complaints about the water that was seen to have a foul odour taste or colour. For example, in 2019, Air Selangor Sdn Bhd, the water utility company supplying water to Selangor and her neighbouring regions, received a whopping 1,143 complaints before any action was taken on the bad quality of water [12].

C. Impact of Bad Quality of Tap Water

Many health risks come with heavy metals and other pollutants in the water. For example, a study by the Environmental Working group estimated 100,000 cancer-causing cases might be related to tap water. They also proposed the national regulations need to keep up with the calibre of science today. Other than that, heavy metals, namely Ph, have detrimental effects on kids, giving them mental disabilities and many other sicknesses. In Flint, Michigan, we saw this in the United States of America; it affected a whole new generation of children in that town [13].

In Malaysia, organisations like WHO and DOE stated that it is good practice for consumers to install private water filters that are installed directly to the main water supply to mitigate the heavy metals if present, but based on a study of The Usage of Home Water Filtration System in Malaysia they found that some water filters are better than others, they also found water filters that did not work as well as they were advertised to [12].

IV. SIMILAR SYSTEMS

In this section, a discussion on the similar available systems for users will be looked at and compared to the proposed approach. Many Do It Yourself (DIY) systems need to be put together by users and almost no similar competitors that sell to consumers. Moreover, none of the products is made for the Malaysian consumer market to be installed and monitored in real-time [15].

A. Libelium Smart Water Xtreme Sensors

Libelium is an award-winning company that focuses on designing and manufacturing technology solutions based on IoT technology. They have created a water management system that monitors the quality of water with the use of IoT. Their system is called the Smart Water Xtreme Sensors, shown in Fig. 3.

This product is made for industrial uses as well as emergency, agriculture, and consumers. This system will continuously monitor the water pH level, suspended solids, ammonium, nitrates and phosphorus, conductivity and BOD

and COD in the water. The design of the system focuses on industrial use like sea and lake installations rather than consumer use. This system boasts low power consumption, robust sensor probes. There is also a model that incorporates solar power and GPS [16].

For the set-up and collecting of data from the system, a computer is needed. The user would also have some programming knowledge shown in Fig. 3. The system's price is a staggering €10,950.00, which in Malaysia would cost more than RM 50,000[16]. Furthermore, this system does not come with a mobile application, and after a user collects 100 water samples, the user would need to pay a subscription fee based on the usage of the database [16].



Fig. 3. Set up of Libelium Smart Water Xtreme Sensors [16]

In addition, the user is expected to clean the sensors every two weeks and ensure the system is not damaged due to too much exposure to the elements [16].

B. Winmore Online Water Quality Sensor

Yantai Winmore is a company that focuses on solutions for industrial applications, and therefore their system is concentrated in an industrial environment. The system has interchangeable sensors, and up to 6 sensors can be used simultaneously, which monitor the oxygen level in the water, pH, conductivity, and turbidity. In addition, they also boast special sensors such as aquaculture special dissolved oxygen sensor and industrial oxygen sensor for wastewater treatment [17].

The sensors in this product stores data and is equipped with an auto-cleaning device. This device is meant to be submersible and only taken out. It boasts real-time monitoring but is also only accessible through a laptop. The device can encapsulate up to 6 sensors in an integrated design, as illustrated in Fig. 4 and Fig. 5 [17].



Fig. 4. Winmore Online Water Quality Sensor [17]



Fig. 5. Winmore Online Water Quality Sensor with sensors connected [17]

In order to get more information about this product, a business-to-business representative is needed as this company does not sell it to the public and only focuses on industrial sales.

TABLE 1: COMPARISON OF COMPETITORS

Monitoring system Features	Libelium Smart Water Xtreme Sensors	Winmore Online Water Quality Sensor
Real-time monitoring	Available	Available
Mobile Application	Unavailable	Unavailable
Maintenance	High	Low
History Recording	Available	Available
Device needed	Computer	Computer
Consumer base	Industrial	Industrial
User-friendliness	Yes, but it requires basic programming or technical knowledge.	Unknown
pH sensor	Yes	Yes
Turbidity Sensor	Yes	Yes
O2 Sensor	Yes	Yes
Heavy Metals sensor	Yes	Yes
Database	Yes, but with a subscription fee	Unknown
Connectivity	Zigbee	Unknown
Waterproof	Partially	Yes
Available to public	No	No, only for industrial use
Home environment usage	Only for the swimming pool	No
Encapsulated	2 Separate encapsulations	One single encapsulation
Affordability	Very expensive	Unknown

This study has provided new insight and a clearer understanding of the proposed system and the factors that should be considered to provide the public with an efficient and well-designed system to monitor water quality in real-time with ease. Furthermore, after looking at similar systems and realising there are not many solutions for this problem, there is an empty void that should be filled, especially with the data provided about the pollution that makes the proposed project more critical. The researcher also realised that maintenance of the system should be considered and will find

a solution to lower the cost spent on maintenance, be it the time the user spends or the money. With that, based on the research conducted, the proposed system will be a good solution based on the problems with the use of technology.

V. TECHNICAL RESEARCH AND PROPOSED DESIGN

The technical research involves researching the suitability of programming languages, Interactive Development Environment (IDE), database management system, and hardware available to develop the proposed system.

TABLE 2: COMPARISON OF SUITABLE DBMS AND PROGRAMMING LANGUAGE

DBMS	Mongo Database	Firebase Realtime Database
Database Model	Document Store	Document Store
Developer	MongoDB, Inc	Google
Licensing	Open-Source	Commercial
Supported Operating System	Hosted, therefore it is able to be supported on all operating systems	MacOS, Windows, Linux, Solaris
APIs	Android, iOS, JavaScript API, RESTful HTTP API	Proprietary protocols adopting JSON
Programming Language	C, C#, C++, Java, JavaScript, PHP, Ruby and many more	Java, JavaScript, Objective-C
Security	Secured	Secured
Suitable project size	Small to medium	Medium to huge
Suitability	For desktop and application- based database	For desktop and application- based database

Based on the comparison in Table 2, Firebase Realtime Database seems to be a suitable database to manage the data of the proposed system as it has a slight edge in the ease-of-use department. This is because the database itself is easily understood and not as complex as the Mongo database. And also, since the Firebase database is an amicable cross-platform system, it is more suited to the proposed project as it will enable the developers to use the same database for the Android application easily. Moreover, the Firebase database has a free version and extensive documentation on the database itself, which will help in the development process.

For the hardware, the user chose to use the Arduino Uno as it is cheap and easily accessible and is the most popular of all Arduino microcontrollers, which means there is a lot of information regarding this microcontroller which will help the developers have more resources to see in the case there is a problem with the hardware. The Arduino Uno is depicted in Fig. 6.

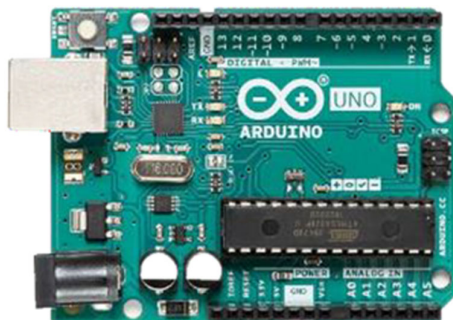


Fig. 6. Arduino Uno Microcontroller

To connect the system to the internet, the researchers chose an ESP8266 NodeMCU, which can easily be programmed on the Arduino IDE with the same knowledge and a few libraries and board changes. The specific NodeMCU used is the LoLin NodeMCU V3; this board can connect to WiFi services only and is depicted in Fig. 7.

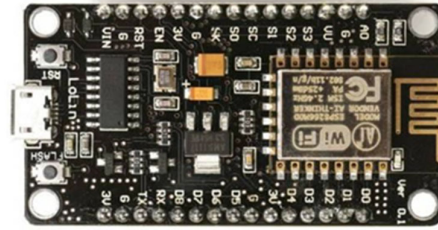


Fig. 7. NodeMCU LoLin V3

The pH sensor can detect the pH level in the water. It is important because an imbalance in pH water level will affect the body's metabolic process, which will lead to many health complications, including skin problems, slow immune systems, and the inability to absorb nutrients in the form of vitamins. All this can be detrimental to the human body. The probe and module chosen are depicted in Fig. 8.



Fig. 8. PH4502 and pH probe

Both the probe and module are made by DIY More, a reputable manufacturer of robotics. The specific module chosen is the PH4502 pH meter, and the probe is a generic probe that the company manufactures.

The other sensor chosen by the researchers is the sensor that measures the Total Dissolved Solids (TDS). The TDS sensor works by measuring the conductivity within the water and values it in particles per million which can also be expressed as mg/L (milligram per litre). Therefore, the perfect reading would be 0 as this would mean there are completely no dissolved solids which indicate the water is not contaminated, as shown in Fig. 9.

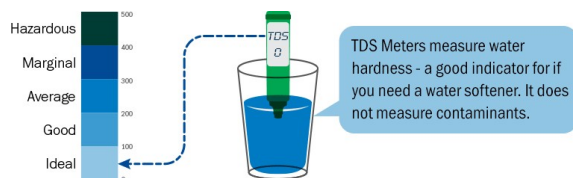


Fig. 9. perfect TDS reading

TDS sensors cannot distinguish the metals or solids in water, but a high reading would mean detrimental consequences either way, and this would indicate that further testing should be taken; therefore, this sensor makes more sense; hence the researchers chose this sensor. The TDS sensor also comes with a probe, and the researchers determined the module and probe that Keyestudio makes, specifically their KS0429 essential studio TDS Meter V1.0. Shown in Fig. 10 [20].

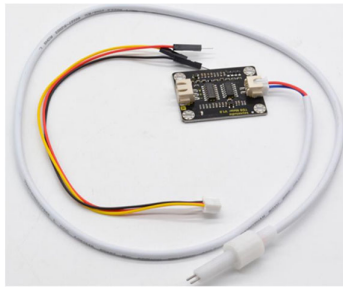


Fig. 10. TDS probe and KS0429 module

The basic development of the system was developed as discussed in the technical research and was achieved through the following points, as illustrated in Fig. 11 and Fig. 12.

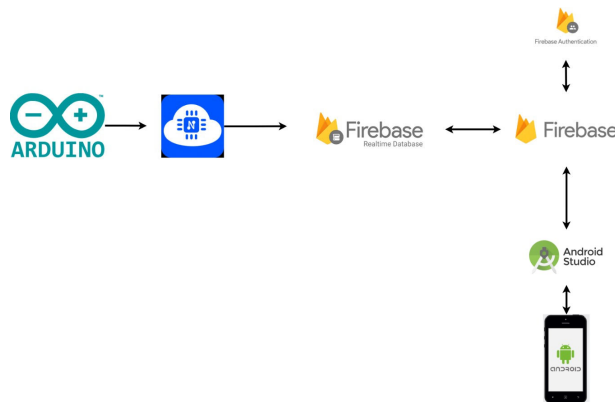


Fig. 11. IoT Based Smart Water Quality Monitoring System Architecture Design

- The use of Arduino Uno to collect the data read by the sensors.
- The NodeMCU ESP8266 to receive data from Arduino Uno and send it to Firebase Real-Time Database via a WiFi connection. The use of a TDS sensor and a pH sensor will indicate if the water is healthy or polluted.
- The use of Arduino IDE, Firebase, and Android Studio for the programming of the hardware, database, and android application.

The Smart Water Monitoring System's intended functionality is to be effective and efficient in monitoring the water quality, which means that the hardware components used need to communicate in harmony with each other and with the software that is being used. This ensures that readings from the sensor are sent to the database are accurate and updated in real-time, which then is extracted by the android

application. The dependent functionalities are as listed in the following pattern:

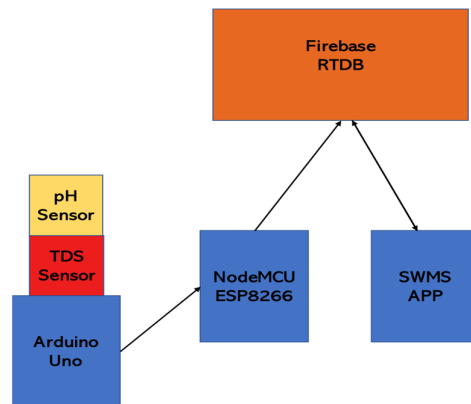


Fig. 12. Overall illustration of the system

- Users will need to register on the application with an email and password if they have not. Upon registering, the user email and password will be stored and validated by Firebase Sign-in Authentication.
- Login users will be directed to the main page of the application, which will display the real-time data being read by the sensors.
- Users will have tabs within the app for different functions, one of which will be to send an email to the user's email with the history of the readings; another option would be to send the data to the utility company.
- There is also a button for logout on the main screen.

A. Abstract Architecture

As shown in Fig. 13, the system has six main components: Arduino, NodeMCU, Firebase Real-time Database, Firebase Authenticator, and Android Studio. The system starts as readings from the hardware side, specifically the Arduino Uno microcontroller connected to NodeMCU; the sensor data from Arduino gets sent to the NodeMCU connected to Firebase via WiFi connection. The NodeMCU contains the secret key and firebase host, connecting to the system and adding data. The sensor data collected by the NodeMCU is then set and pushed into Firebase Realtime Database.

The other component in the system is the android application that is also connected to the Firebase Real-time database, which allows the application to retrieve the sensor data that is updated in real-time by the NodeMCU and display it within the mobile application. On top of that, Android Studio also uses another Firebase service, the Firebase Authenticator that allows the application to have a feature that registers users and authenticates their email and password during login. The Real-time database is also used to show historical sensor data displayed in RecyclerView of Android Studio that helps the application work smoothly and use fewer resources when accessed. There are a total of 4 main children in the Firebase Realtime Database. The application development was done with Android Studio's; the developed system is only on the android platform; therefore, the only IDE used to display the data is Android Studio.

B. System Circuit Design

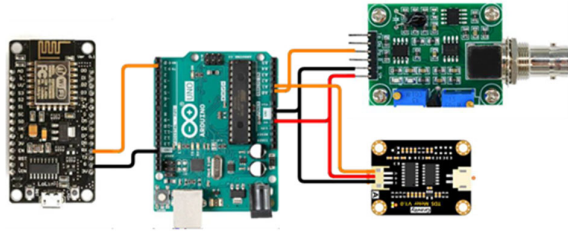


Fig. 13. Circuit Diagram of the Sensors

In Fig. 13, the depiction of the circuit diagram is shown. In the middle is the Arduino Uno module to the left is the NodeMCU ESP8266 module's ground which is connected to the Uno's ground which is the black cable, and the RX pin of the NodeMCU is connected to the TX pin of the Uno, which gives the communication of the two modules. On the top right side is the DIY More's PH4502 chip. This chip connects to the pH probe and conveys the readings to the Uno. In this chip connected to the Uno via ground, 5V and A0 serial port, unlike the NodeMCU, which needs an individual power source, the PH4502 module gets its power from Arduino. Similarly, the KS0429 key studio TDS Meter V1.0 also requires power from the Uno and is also connected to the ground and A1 Serial Port of the Arduino. The serial ports are the ones that are receiving information from these chips, whereas the Uno via the TX port is sending the data collected to the NodeMCU, which receives it via its RX pin. The NodeMCU is responsible for connecting this whole system to the firebase database with WiFi. In Table 3, the sensor modules share a 5V port and a Ground port but have individual Serial Ports.

TABLE 3: CIRCUIT CONNECTIONS BETWEEN NODEMCU AND ARDUINO

Arduino Uno	NodeMCU	Cable colour
Ground (GND)	Ground (G)	Black
TX	RX	Orange

Arduino Uno	PH Meter (PH4502)	TDS Meter (KS0429 V1.0)	Cable colour
Ground (GND)	Ground (GND)	Ground (-ve)	Black
5V	V+	+	Red
Serial Port A0	Po	Not connected	Yellow
Serial Port A1	Not connected	A0	Yellow

C. System Implementation



Fig. 14. System in a weather resistant junction box

The system with the same connections is placed in an enclosure with two holes to allow power into the box and allow the sensor's wire to go into the box and connect to its modules. This enclosure will keep the electronics safe from the heat and the rain when it is closed.

This enclosure is a SIRIM approved water filter housing that can withstand the pressure of the water from the main pipes and also withstand the elements from heat to rain; it also enables water to pass through freely while enabling the sensors to get samples at the same time protecting the sensors from getting damaged.



Fig 15. Water Filter Enclosure with pH Probe and TDS Probe Sensors

D. Interface Design

In the screenshots of the actual android application that has been developed, it must be noted that the layout of the applications may be altered. This is to adapt with phones that have older versions of firmware.

Fig. 16 shows the application's landing page, which will be seen when a user launches the Smart Water Quality Monitoring System or SWQMS. In short, this abbreviation is also featured in the logo. The design of this page has a simple UI and is heavily influenced by water. When any of the buttons on this page are clicked, the user will immediately be directed to the respective pages of their choice

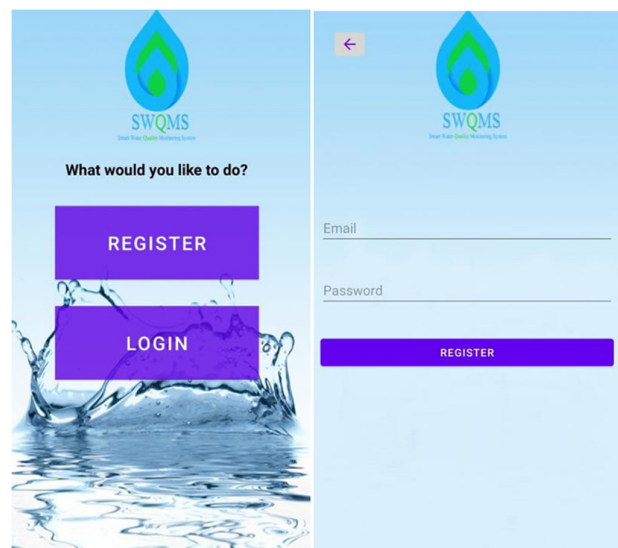


Fig. 16. Launch, Registration and Login Page

The registration page will be directed to if they click on the registration button; it has two empty fields for users to fill in to create their account. The email and password entered here will be verified first to see if they match the criteria of being valid email whereby there is "@" and ".net, .com, etc." and the password will also be verified if it meets the criteria of being at least six characters long. These two fields will then be saved into Firebase Authentication Services. The UI on this page is simple user friendly, and clean to make the user feel fresh and make the page feel uncluttered.

The login page that users will be directed to if they click on the login button has two empty fields for users to fill in to create their account. The email and password entered here will be verified and matching these accounts registered in Firebase Authenticator. The UI on this page is simple user friendly, and clean to make the user feel fresh and make the page feel uncluttered.

In Fig. 17, the homepage can be seen. This page appears when the user has successfully logged in. The user will see the TDS and PH reading in Realtime the readings are on Card View. This UI feature is available in Android Studio, and it gives emphasis on these two fields.

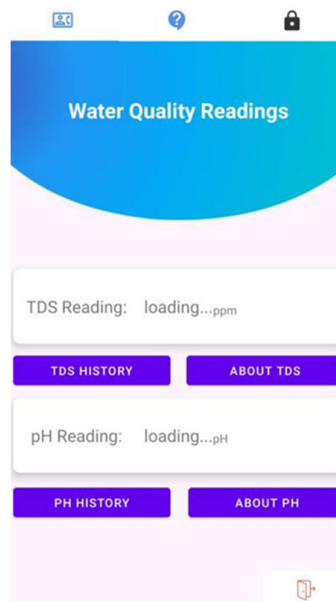


Fig 17: Homepage/ Dashboard

Other than that, the dashboard also has many functionality buttons which were made using a mix of standard controls and image buttons. From the top left, the first button will direct the user to the Utility Contact information page. Next, the question mark bubble button leads users to the admin contact information page, followed by the change password button. The next button is located right below the Card View of TDS readings which takes the user to the TDS history page shown in Fig. 18. Next, the about TDS button takes the user to the About TDS page shown in Fig. 19. These buttons are also similar to the pH Reading Card View. Finally, the about pH button directs the user to the About pH page shown in Fig. 20. The researchers took a minimalist route while designing this UI, making the dashboard look fresh and decluttered. The buttons used are also easily understandable, which leans this application towards a more user-friendly application.

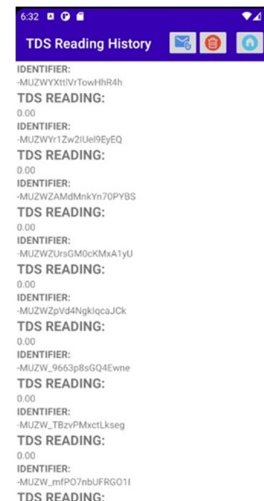


Fig. 18. pH Reading History and TDS Reading

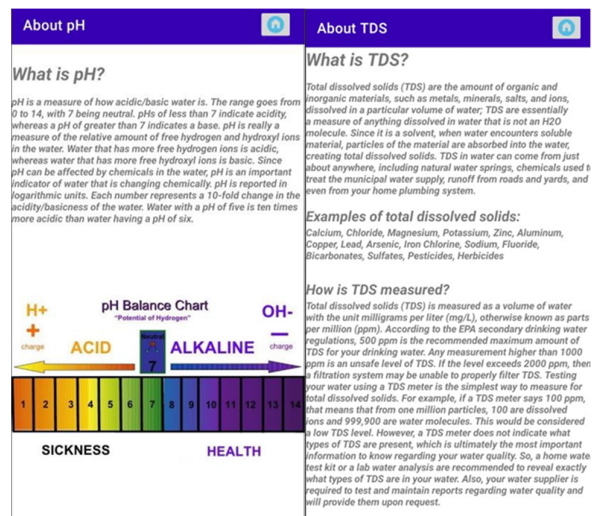


Fig.19. About PH and TDS pages.

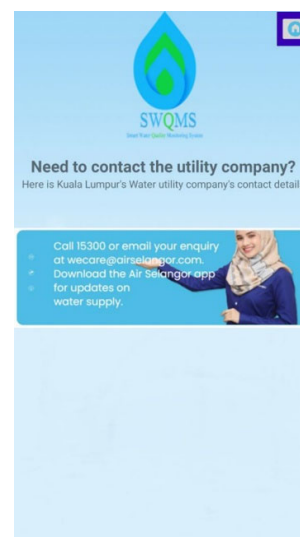


Fig. 20. Contact Admin and Contact utility company's page

VI. CONCLUSION

The first and obvious benefit to this system is monitoring the quality of water that is being supplied to the house. Other than that, people can also take necessary action if fluctuations are observed as utility companies tend to service a vast area, resulting in neglecting some locations as no complaints regarding the water have been made. Therefore this system will give users to act as a check and balance and also be able to notify the utility companies and or authorities if discrepancies are observed. The research carried out by the authors is limited to design only; however, the authors aim to further enhance this research in the future by providing implementation of the design and providing the performance comparison with the competitors. Furthermore, the implementation of this system is pretty broad users will be able to upgrade the system on their own. For instance, if users do not have a WiFi signal near the water meter, they can buy a GPRS module and fit it in. This system can also be used in other piping environments and does not necessarily need to be plugged into the main pipes. Lastly, the information and the functionalities provided within the system were designed with the user in mind. Therefore, the UI is highly user-friendly and interactive, which would mean higher usability for the target users. With that being said it is no doubt that exemplary implementation of the system would bring the system great success. However, some issues would still need to address primarily the scalability, which will also be another factor of the system's success.

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